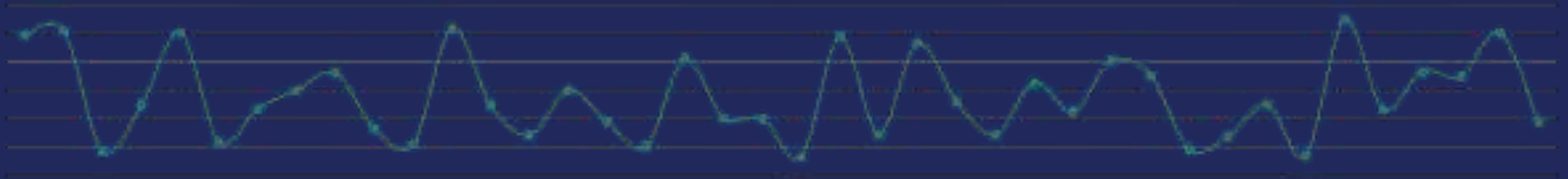


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Time Series Analysis

Session 3 — Moving Average Models & Stationarity Testing



Recall: Stationarity



Weak (covariance) stationarity requires three things to hold over time



Constant mean

The average level doesn't drift up or down over time



Constant variance

The spread of values doesn't grow or shrink over time



Lag-dependent covariance

Covariance between Y_t and Y_{t-k} depends only on k , not on t

Why it matters: most classical time series models assume stationarity. Non-stationary series are typically transformed before modeling.

From Eyeballing to Testing



Visual inspection is a good start — but it isn't proof

- Looking at a plot for trend or a shifting spread is a useful first pass, but two people can easily disagree on what counts as “no trend.”
- Fitting an ARIMA model to a series that isn't actually stationary can produce false relationships and misleading forecasts.
- What we need instead is a quantitative, repeatable decision rule — a hypothesis test we can run directly on the data.
- That is exactly what unit root tests, like the Augmented Dickey-Fuller (ADF) test, provide.

AR(1) — Autoregressive Model of Order 1

$$X_t = \delta + \varphi X_{t-1} + W_t$$

ϕ controls how much the series “remembers” its own past:

$$|\varphi| < 1$$

Stationary — shocks fade out over time

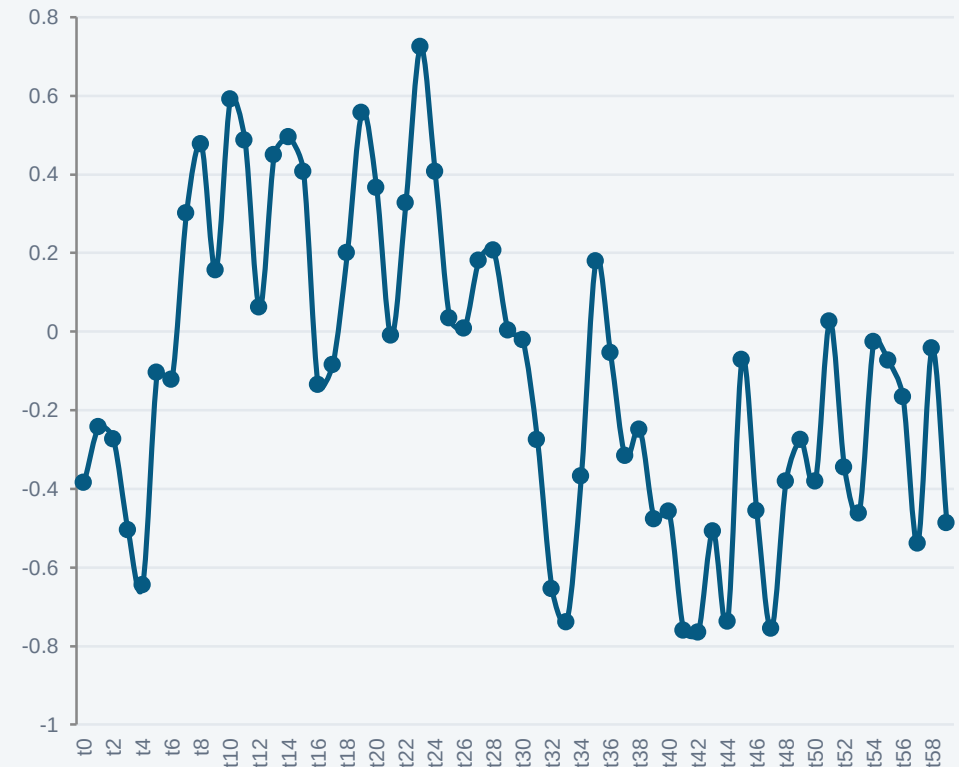
$$\varphi = 1$$

Random walk — non-stationary

$$|\varphi| > 1$$

Explosive — shocks grow without bound

A STATIONARY AR(1) PATH ($\phi \approx 0.7$)

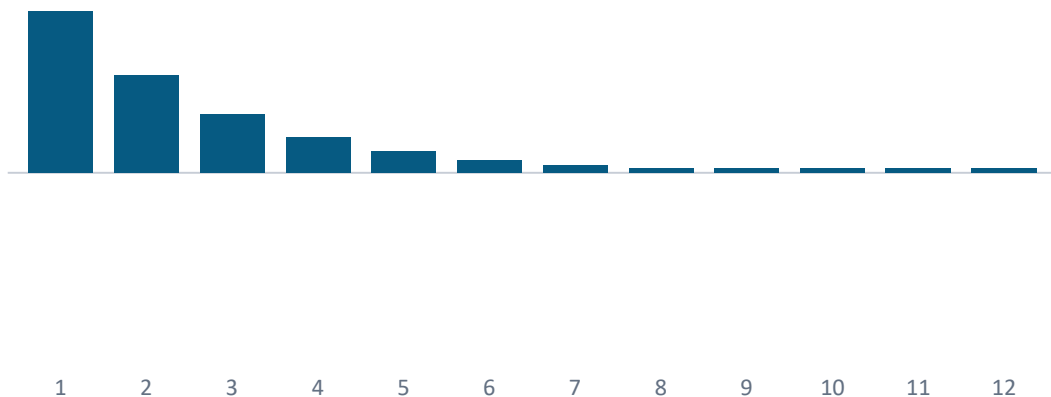


Pattern of the ACF for AR(1)



The sign of ϕ_1 determines how the correlogram decays

$\phi_1 = 0.6$ (positive)



$\phi_1 = -0.7$ (negative)



Why it matters: Positive ϕ_1 gives an ACF that tapers smoothly to 0, all on one side. Negative ϕ_1 gives an ACF that also tapers to 0, but alternates sign at every lag.

Unit Roots & Non-Stationarity



Recall from Session 2 — the boundary case of the AR(1) model

1

The random-walk case

When $\phi_1 = 1$ in $x_t = \delta + \phi_1 x_{t-1} + w_t$, the model becomes a random walk: $x_t = \delta + x_{t-1} + w_t$.

2

“A unit root”

$\phi_1 = 1$ is called a unit root — it is the root of the AR characteristic equation.

3

Why it breaks stationarity

A unit root means variance grows with t and shocks never fade — the series is non-stationary by definition.

Why it matters: A unit root test checks exactly this: does the data behave as if $\phi_1 = 1$?

The ADF Test: Hypotheses



Augmented Dickey Fuller test. A regression-based test for the presence of a unit root

$$\Delta x_t = \alpha + \beta t + \gamma x_{t-1} + w_t$$

1

Null hypothesis (H_0)

$\gamma = 0$ — the series has a unit root and is non-stationary.

2

Alternative hypothesis (H_1)

$\gamma < 0$ — the series is stationary (around a possible trend).

3

A non-standard distribution

The statistic for γ doesn't follow a t-distribution — it follows the Dickey-Fuller distribution, so we need special critical values.

Why it matters: “Augmented” means lagged difference terms $\sum \delta_i \Delta x_{t-i}$ are added to the regression to remove any remaining autocorrelation in the residuals.

ADF vs. KPSS: Complementary Tests



ADF Test

- H_0 : the series has a unit root
- Reject $H_0 \Rightarrow$ evidence of stationarity
- Sensitive to how many lag terms are included
- `statsmodels.tsa.stattools.adfuller(x)`

KPSS Test

- H_0 : the series is stationary
- Reject $H_0 \Rightarrow$ evidence of non-stationarity
- The opposite null hypothesis from ADF
- `statsmodels.tsa.stattools.kpss(x)`

The Transformation Workflow



From a raw series to one ready for AR / MA modeling

1

Plot the series

Look for trend, changing variance, seasonality

2

Stabilize variance

Log transform or Box-Cox if spread grows with level

3

Test for a unit root

Run ADF and KPSS together

4

Difference if needed

Regular and/or seasonal differencing

5

Re-test

Confirm ADF + KPSS now agree: stationary

6

Identify order

Read ACF / PACF of the stationary series for p , q

Stabilizing Variance: Log & Box-Cox



The fix for spread that grows with the level of the series

Log transform

$$Y_t = \log(X_t)$$

Common for economic and financial series with multiplicative, not additive, noise. Stabilizes variance but doesn't remove trend by itself — usually paired with differencing after.

Box-Cox transform

$$Y_t = (X_t^\lambda - 1) / \lambda, \lambda \neq 0$$

Generalizes the log transform ($\lambda = 0$ is exactly the log case). Use when a plain log over- or under-corrects the variance — λ is estimated from the data.

Typical order: stabilize variance first (log / Box-Cox), then test and difference — not the other way around.

Removing Stochastic Trends: Differencing



The fix for a unit root — series behaving like a random walk

Regular differencing

$$\nabla X_t = X_t - X_{t-1}$$

Use when ADF fails to reject and the ACF decays very slowly — the classic random-walk signature.

Seasonal differencing

$$\nabla_s X_t = X_t - X_{t-s}$$

Use when ACF shows repeating spikes at lags $s, 2s, \dots$ e.g. $s = 12$ for monthly, yearly seasonality.

Watch for over-differencing

One difference is usually enough. A second difference ($\nabla^2 X_t$) often introduces artificial negative autocorrelation and inflates MA structure that wasn't in the original data.

Choosing the Right Fix

A quick decision guide from symptom to transformation



Symptom	Fix
Slowly-decaying ACF, mean wanders (random-walk-like)	Differencing
Repeating seasonal pattern (spikes at lag s , $2s$, ...)	Seasonal differencing
Variance grows with the level of the series	Log / Box-Cox transform
Both trend and heteroskedastic variance	Log / Box-Cox first, then difference

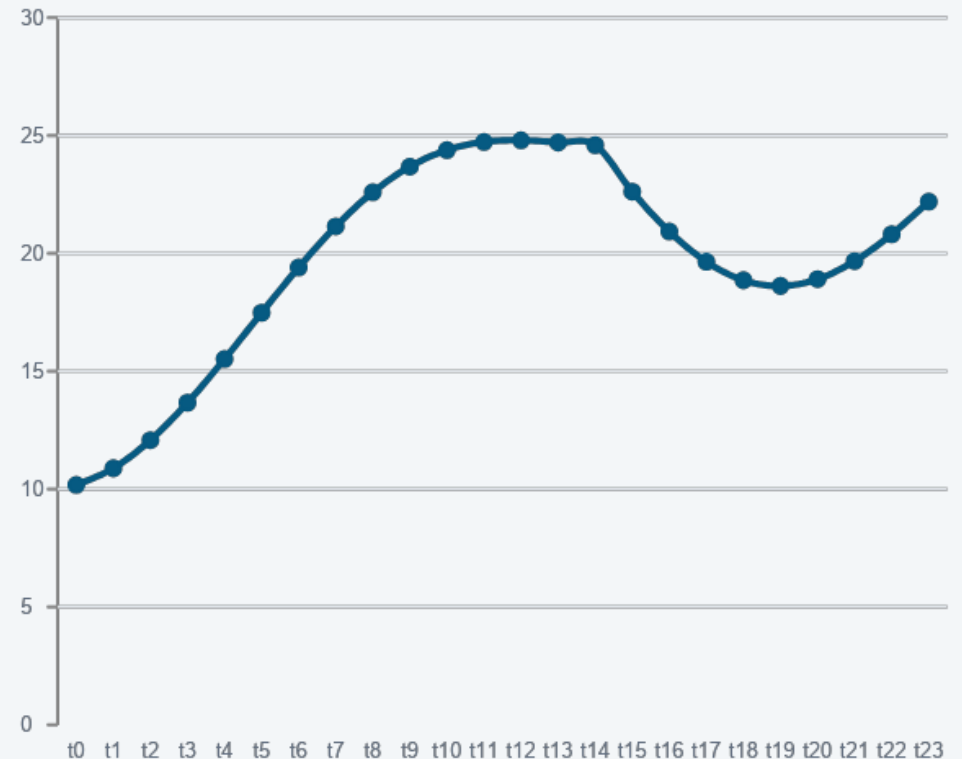
What Is a Moving Average Term?

A moving average term in a time series model is a past forecast error w_{t-1} , multiplied by a coefficient — not a past value of the series itself.

The key difference: AR terms reuse past values of x . MA terms instead reuse past shocks w — a different kind of memory that fades out after a fixed number of lags.

- MA(1): one lagged error term
- MA(2): two lagged error terms
- MA(q): q lagged error terms
- ARMA: AR and MA terms combined

A SIMULATED MA(1) SERIES



The MA(1) Model



Moving average model of order 1 — today's value depends on today's and one past shock

$$x_t = \mu + w_t + \theta_1 w_{t-1}$$

1

White noise errors

w_t is independent and identically distributed with mean 0 and constant variance σ_w^2 .

2

One-lag memory

Only yesterday's shock w_{t-1} feeds into today's value — earlier shocks have already dropped out.

3

Invertibility needs $|\theta_1| < 1$

This ensures the MA(1) has an equivalent, converging infinite-order AR representation.

Why it matters: Unlike AR(1), MA(1) is stationary for any value of θ_1 — its invertibility, not stationarity, that constrains θ_1 .

Properties of the MA(1) Model



Mean, variance, and autocorrelation follow directly from the model equation

μ

Mean: $E(x_t) = \mu$

σ^2

Variance: $Var(x_t) = \sigma_w^2(1 + \theta_1^2)$

ρ

Autocorrelation: $\rho_1 = \frac{\theta_1}{1+\theta_1^2}$ and $\rho_h = 0$ for all $h \geq 2$

Only lag 1 is nonzero — every higher lag has $\rho_h = 0$

Why it matters: A single nonzero autocorrelation at lag 1, followed by a hard cutoff to zero, is the fingerprint of an MA(1) process in a sample ACF.

The MA(2) and General MA(q) Model



Adding more lagged shocks extends the memory of the model

$$x_t = \mu + w_t + \theta_1 w_{t-1} + \theta_2 w_{t-2} + \dots + \theta_q w_{t-q}$$

1

Variance grows with each term

$\text{Var}(x_t) = \sigma^2 w(1 + \theta_1^2 + \theta_2^2 + \dots + \theta_q^2)$ — every added term adds variance.

2

ACF is nonzero for lags 1..q

Each of the first q lags can have a nonzero autocorrelation, shaped by the θ coefficients.

3

ACF is exactly 0 beyond lag q

This is the defining signature of an MA(q): a hard cutoff, not a gradual decay.

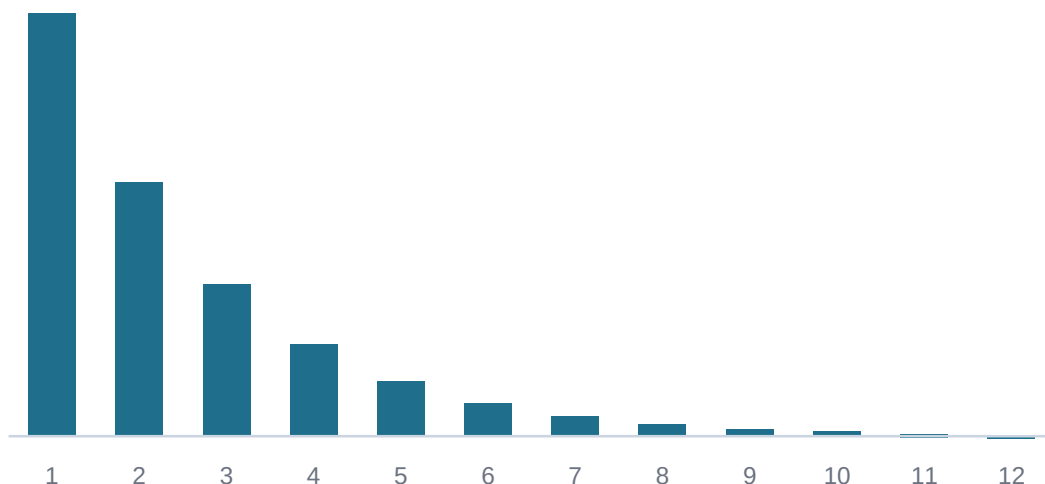
Why it matters: A sample ACF with significant spikes at lags 1 and 2 only (and nothing beyond) is a strong signal for an MA(2).

ACF Decay (AR) vs. ACF Cutoff (MA)

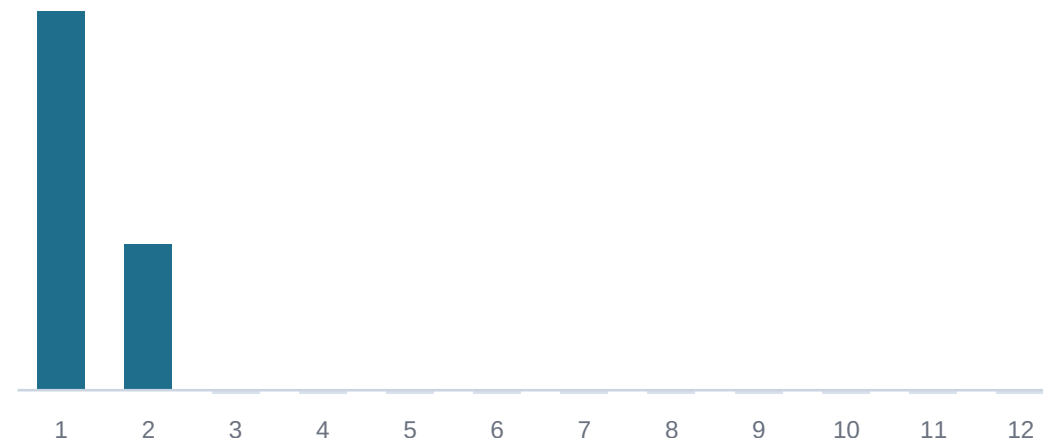


A quick visual test for telling the two model families apart

AR(1), $\phi_1 = 0.6$ — decays gradually, never truly hits zero



MA(2), $\theta_1 = 0.6, \theta_2 = 0.3$ — sharp cutoff after lag 2



Why it matters: An AR(p) ACF tapers off but never exactly reaches zero — it keeps a faint signature at every lag. An MA(q) ACF is the opposite: nonzero for lags 1...q, then exactly zero from lag q+1 onward. Gradual tapering says “AR”; a sharp cutoff says “MA.”

Identifying AR vs. MA: ACF & PACF Together



AR(p) model

- PACF shuts off after lag p
- ACF decays gradually (tapers)
- Best identified from the PACF
- Order p = last significant PACF spike

MA(q) model

- ACF shuts off after lag q
- PACF decays gradually (tapers)
- Best identified from the ACF
- Order q = last significant ACF spike